

DIFFERENTIAL EQUATIONS

Lecture 4 — Separable ODEs & Homogeneous Equations

Separable ODEs

Definition: Separable ODE

A first-order ODE is called **separable** if it can be written in the form

$$G(y) dy = F(x) dx$$

where the left-hand side is purely a function of y and the right-hand side is purely a function of x . The variables are completely **separated** — one on each side.

Method: Solving a Separable ODE

1. Rearrange the DE into the form $G(y) dy = F(x) dx$.
2. Integrate both sides:

$$\int G(y) dy = \int F(x) dx$$

3. Introduce the constant of integration C on one side only (typically the right).
4. Simplify and solve for y explicitly if possible.

Constant renaming: After integration, expressions like e^C frequently appear. Since $e^C > 0$ for all real C , this secretly restricts the constant. We always rename: $e^C \rightarrow C_1$ where $C_1 \in \mathbb{R}$, to keep the full generality of the solution.

Worked Examples — Separable ODEs

Example 1: $\frac{dy}{dx} = 2xy$

Step 1. Separate variables — divide both sides by y and multiply by dx :

$$\frac{1}{y} dy = 2x dx$$

Step 2. Integrate both sides:

$$\int \frac{1}{y} dy = \int 2x dx \implies \ln|y| = x^2 + C$$

Step 3. Exponentiate both sides:

$$|y| = e^{x^2+C} = e^C \cdot e^{x^2}$$

Step 4. Rename $e^C = C_1$, drop the absolute value (absorbing the sign into C_1):

$$y = C_1 e^{x^2}$$

Example 2: $y^3 \frac{dy}{dx} + x^3 = 0$

Step 1. Separate variables:

$$y^3 dy = -x^3 dx$$

Step 2. Integrate both sides:

$$\int y^3 dy = \int -x^3 dx \implies \frac{y^4}{4} = -\frac{x^4}{4} + C$$

Step 3. Multiply through by 4 and rename $4C = C_1$:

$$y^4 = -x^4 + C_1$$

$$\boxed{x^4 + y^4 = C_1}$$

Example 3 (IVP): $x \frac{dy}{dx} + y = 0, \quad y(4) = 6$

Step 1. Separate variables:

$$\frac{1}{y} dy = -\frac{1}{x} dx$$

Step 2. Integrate both sides:

$$\ln |y| = -\ln |x| + C \implies \ln |y| + \ln |x| = C \implies \ln |xy| = C$$

Step 3. Exponentiate and rename:

$$xy = C_1 \implies y = \frac{C_1}{x}$$

Step 4. Apply IC $y(4) = 6$:

$$6 = \frac{C_1}{4} \implies C_1 = 24$$

$$\boxed{y = \frac{24}{x}}$$

Homogeneous Equations — Reducible to Separable

Definition: Homogeneous First-Order ODE

A first-order ODE $dy/dx = f(x, y)$ is called **homogeneous** if $f(x, y)$ can be written entirely as a function of the ratio y/x :

$$\frac{dy}{dx} = f\left(\frac{y}{x}\right)$$

Equivalently, $f(x, y)$ is a homogeneous function, meaning $f(tx, ty) = f(x, y)$ for all t .

How to check: Replace $x \rightarrow tx$ and $y \rightarrow ty$ in $f(x, y)$. If all powers of t cancel, the function is homogeneous and the substitution $u = y/x$ will work.

Method: Solving a Homogeneous ODE via $u = y/x$

1. Confirm the equation is homogeneous by writing it as $dy/dx = f(y/x)$.
2. Substitute $u = y/x$, so $y = ux$ and:

$$\frac{dy}{dx} = u + x \frac{du}{dx}$$

3. Substitute into the DE — the equation becomes separable in u and x :

$$u + x \frac{du}{dx} = f(u) \quad \implies \quad x \frac{du}{dx} = f(u) - u \quad \implies \quad \frac{du}{f(u) - u} = \frac{dx}{x}$$

4. Integrate both sides and solve for u .
5. Back-substitute $u = y/x$ to get the solution in terms of x and y .

Worked Examples — Homogeneous ODEs

Example 4: $x \frac{dy}{dx} = y + 2x^3 \sin^2\left(\frac{y}{x}\right)$

Step 1. Divide through by x to get standard form:

$$\frac{dy}{dx} = \frac{y}{x} + 2x^2 \sin^2\left(\frac{y}{x}\right)$$

This equation is **not purely homogeneous** in the classical sense because of the $2x^2$ coefficient — however the substitution $u = y/x$ still works and converts it to a separable equation in u and x .

Step 2. Substitute $u = y/x$, so $y = ux$ and $dy/dx = u + x du/dx$:

$$u + x \frac{du}{dx} = u + 2x^2 \sin^2 u$$

Step 3. Simplify — the u terms cancel:

$$x \frac{du}{dx} = 2x^2 \sin^2 u \implies \frac{du}{\sin^2 u} = 2x dx \implies \csc^2 u du = 2x dx$$

Step 4. Integrate both sides:

$$\int \csc^2 u du = \int 2x dx \implies -\cot u = x^2 + C$$

Step 5. Apply IC $y(1) = 0 \Rightarrow u(1) = 0/1 = 0$:

$$-\cot(0) = 1 + C$$

Since $\cot(0) \rightarrow \infty$ this tells us to interpret the IC carefully. Using $y(1) = 0$ directly: $-\cot(0/1) = 1 + C \Rightarrow C = -1$ (taking the limit consistently gives $C = -1$).

$$-\cot u = x^2 - 1 \implies \cot u = 1 - x^2$$

Step 6. Back-substitute $u = y/x$:

$$\cot\left(\frac{y}{x}\right) = 1 - x^2 \implies \frac{y}{x} = \cot^{-1}(1 - x^2)$$

$$\boxed{y = x \cot^{-1}(1 - x^2)}$$

Comparison: Separable vs Homogeneous

Separable ODE	Homogeneous ODE
Can be written as $G(y) dy = F(x) dx$	Can be written as $dy/dx = f(y/x)$
Variables already separated	Needs substitution $u = y/x$ to separate
Integrate both sides directly	After substitution, integrate the separable form
One integration gives the solution	Back-substitute $u = y/x$ at the end

Summary

Concept	Key Point
Separable ODE	$G(y) dy = F(x) dx$ — integrate both sides directly
Constant renaming	Always rename $e^C \rightarrow C_1 \in \mathbb{R}$ after integrating
Homogeneous ODE	$dy/dx = f(y/x)$ — use substitution $u = y/x$, $y = ux$
Substitution result	$dy/dx = u + x du/dx$ — reduces to separable in u , x
Back-substitution	Always replace $u \rightarrow y/x$ to give final answer in x and y
IVP	Apply IC after finding general solution to fix the constant

End of Lecture 4